

Unit-3

Descriptive Statics

↳ Data Summarization

↳ Mean, mode, Median & Range

Problem-1

Marks of 10 Students are:

45, 52, 60, 48, 75, 60, 55, 68, 72, 50

Step ①

$$\text{mean} \Rightarrow \frac{\sum x}{n} = \frac{45 + 52 + 60 + 48 + 75 + 60 + 55 + 68 + 72 + 50}{10} = \frac{585}{10} = 58.5$$

Step ②

$$\text{Median} \Rightarrow 45, 48, 50, 52, 55, 60, 60, 68, 72, 75$$
$$= \frac{55 + 60}{2} = 57.5$$

Step ③

mode $\Rightarrow$ 60 occurs twice other occurs only once.

Step 501:-

Measure	Value
mean	58.5
median	57.5
mode	60
Range	30

7 Data Distribution:-

1. Following are the no. of 15 students spent for studying 2, 4, 3, 5, 2, 8, 4, 3, 5, 7, 2, 4, 6, 5, 3, Find Frequency Distribution

step-1: Identify unique values

The values range from 2 to 7.

step-2: ~~Count~~ Count the frequency

Study Hours Frequency

2	3
3	3
4	3
5	3
6	2
7	1
Total	15

step 3: Calculate relative frequency

For example, for 2 hours:

$$\text{Relative Frequency} = 3/15 = 0.20$$

Therefore, Percentage:

$$0.20 \times 100 = 20\%$$

similarly

Hours	Frequency	Percentage
2	3	20%
3	3	20%
4	3	20%
5	3	20%
6	2	13.33%
7	1	6.67%

Conclusion:-

most students studied between 2 and 5 hours, while only one student studied for 7 hours

3. Data Distribution - Quartiles and IQR

Problem 3

consider the following dataset:

10, 12, 15, 18, 20, 22, 25, 28, 30, 35

Find

* Q_1

* Q_2

* Q_3

* interquartile Range (IQR)

Step 1: Find Q_2

There are 10 observations.

$$Q_2 = 20 + 22/2 \quad Q_2 = 21$$

Step 2: Find Q_1

Lower half:

10, 12, 15, 18, 20

Middle value:

$$Q_1 = 15$$

Step 3: Find Q_3

Upper half:

22, 25, 28, 30, 35

Middle value:

$$Q_3 = 28$$

Step 4: calculate IQR

$$IQR = Q_3 - Q_1 = 28 - 15 = 13$$

4. Outlier Treatment - IQR Method

Problem 4

consider the dataset:

10, 12, 13, 4, 15, 16, 17, 18, 20, 50

Identify the outlier using the IQR method

Step 1: Find Q_1

Lower half:

10, 12, 13, 4, 15

Therefore:

$$Q1 = 13$$

step 2: Find $Q3$

upper half:

16, 17, 18, 20, 50

Therefore:

$$Q3 = 18$$

step 3: Calculate IQR

$$IQR = Q3 - Q1 = 18 - 13 = 5$$

step 4: Calculate lower boundary

$$\text{lower Bound} = Q1 - 1.5(IQR) = 13 - 1.5(5) = 13 - 7.5 = 5.5$$

step 5: Calculate upper boundary

$$\text{upper Bound} = Q3 + 1.5(IQR) = 18 + 1.5(5) = 18 + 7.5 = 25.5$$

step 6: Identify outlier

Any value below 5.5 or above 25.5 is an outlier.

$$50 > 25.5$$

Therefore:

50 is an outlier.

step 7: outlier treatment

Depending of the application, we can:

- * Remove the observation

- * Replace it with median

if median replacement is selected

$$\text{Median} = 215 + 16 = 15.5$$

The treated dataset becomes:

10, 12, 13, 14, 15, 15.5, 16, 17, 18, 20

in a real Data Science project, an outlier should not

automatically be deleted first determine whether it

represents an error or a legitimate extreme observation

5. Outlier Detection - Z-Score Method

Problem 5

Suppose the scores are:

40, 42, 45, 46, 48, 50, 52, 90

Assume:

$$\text{Mean} = 51.625$$

and approximately.

$$SD = 15.69$$

Determine whether 90 is an outlier using the Z-Score.
~~it represents an error or a legitimate extreme observation~~

~~5. outlier Dete~~

Step 1: Formula

$$Z = \frac{X - \mu}{\sigma}$$

For $X = 90$:

$$Z = \frac{15.69 \cdot 90 - 51.62}{15.69} = 2.45$$

Step 2: Apply rule

A commonly used rule is: $|Z| > 3$

$$|2.45| > 3$$

indicates a strong potential outlier

Since:

$$2.45 < 3$$

90 is not an outlier according to the ± 3 rule.

However, depending on the chosen threshold, such as ± 2 , it may be flagged for further investigation.

~~15.69~~

~~15.69~~

Problem 6

Consider:

2, 3, 3, 4, 4, 5, 5, 6, 10

Determine the type of skewness

step 1: Find Mean

$$\text{Mean} = 42/9 = 4.67$$

step 2: Find Median

The middle Value is:

4

step 3: Compare Mean and Median

$$\text{Mean} = 4.67 \quad \text{Median} = 4$$

Therefore:

$$\text{Mean} > \text{Median}$$

This indicates Positive/right skewness.

interpretation

The large value 10 pulls the mean toward the right.

Therefore:

Distribution is Positively skewed

General rule

Relationship

Distribution

Mean $\approx$ median

Approximately Symmetric

Mean $>$ Median

Positively/right skewed

Mean $<$ median

Negatively/left skewed

1. Continuous Distribution - Uniform Distribution

Problem 7

The waiting time for a bus is uniformly distributed between

0 and 20 minutes.

Find the Probability that a Passenger waits between 5 and 12 minutes

step 1: Identify limits

$$a = 0, b = 20$$

We need:

$$P(5 < X < 12)$$

step 2: Use uniform distribution

For a continuous uniform distribution:

$$P(c < X < d) = d - c / b - a$$

Therefore:

$$P(5 < X < 12) = 12 - 5 / 20 - 0 = 7/20$$

$$P = 0.35$$

Therefore, there is a 35% Probability that the Passenger waits between 5 and 12 minutes.

8. Continuous Distribution - Normal Distribution

Problem 8

The heights of students are normally distributed with:

$$\mu = 170 \text{ cm} \quad \sigma = 10 \text{ cm}$$

Find the z-score for a student whose height is 185 cm

Step 1: Formula

$$Z = (X - \mu) / \sigma$$

Step 2: Substitute Values

$$Z = (185 - 170) / 10$$

$$Z = 15 / 10$$

$$Z = 1.5$$

interpretation

The student's height is 1.5 standard deviations above the mean

9. Normal Distribution - Probability

Problem 9

using the previous example, Find the probability that a randomly selected student has a height less than 185 cm.

We already calculated:

$$Z = 1.5$$

From the standard normal table:

$$P(Z < 1.5) \approx 0.9332$$

Therefore:

$$P(X < 185) = 0.9332$$

α

93.32%.

interpretation

Approximately 93.32% of students are expected to have heights below 185 cm.